

CASE STUDY

AS-REP Roasting Exposure Detection & Investigation

Guided SOC case study using official Splunk Attack Data

Category

Splunk · PowerShell · Windows Security · Detection
Engineering

Status

Completed

MITRE ATT&CK

T1558.004 — AS-REP Roasting

Data source

Official Splunk Attack Data — T1558.004 PowerShell
dataset

Overview

A guided SOC case study using official Splunk Attack Data. The investigation reviewed PowerShell Script Block Logging and Windows Security events, identified a Kerberos pre-authentication configuration change, built a focused SPL detection, and saved it as a scheduled alert inside Splunk Enterprise.

Key evidence

- PowerShell Event ID 4104 exposed ScriptBlockText.
- The logged command was: `Get-ADUser Guest | Set-ADAccountControl - DoesNotRequirePreAuth:$true`
- Windows Security data contained Event ID 4738 account-change events.
- A focused SPL detection searched for DoesNotRequirePreAuth in PowerShell ScriptBlockText.
- The detection returned two matching events.
- A scheduled alert named AS-REP PreAuth Modification Detection was created and enabled.

PowerShell evidence

PowerShell Event ID 4104 exposed ScriptBlockText and confirmed the presence of a command that sets DoesNotRequirePreAuth to true for the Guest account. This demonstrates a Kerberos pre-authentication configuration change inside the training dataset.

Windows Security evidence and correlation limit

Windows Security Event ID 4738 was reviewed as supporting account-change evidence.

However, this case study does not claim that the reviewed event for LYNETTE_BLANCHARD was proven to be the direct result of the Guest PowerShell command, because that direct correlation was not established.

Detection logic

A focused SPL search extracted ScriptBlockText from the raw event data and searched for the DoesNotRequirePreAuth pattern so the finding could be operationalized as a scheduled alert in Splunk.

SPL Query

```
index=main host="asrep-lab" sourcetype="attackdata:powershell:xml"
| rex field=_raw "(?s) (?.*?)"
| search ScriptBlockText="*DoesNotRequirePreAuth*"
| table _time ScriptBlockText
```

What I learned

Basic Splunk search workflow
Filtering by index, host, and sourcetype
Using rex to extract fields from raw XML
PowerShell Event ID 4104
Windows Security Event ID 4738
search / table / stats
Scheduled alert
Evidence vs assumptions

Important limitations

- Do not claim that credentials were stolen, a password hash was cracked, or a successful compromise occurred.
- Do not state that the reviewed Event ID 4738 for LYNETTE_BLANCHARD was proven to be the direct result of the Guest PowerShell command.
- The alert detects a configuration-change pattern; it is not proof of a completed AS-REP roasting attack by itself.

Selected evidence

Event 4104 exposed the executed script



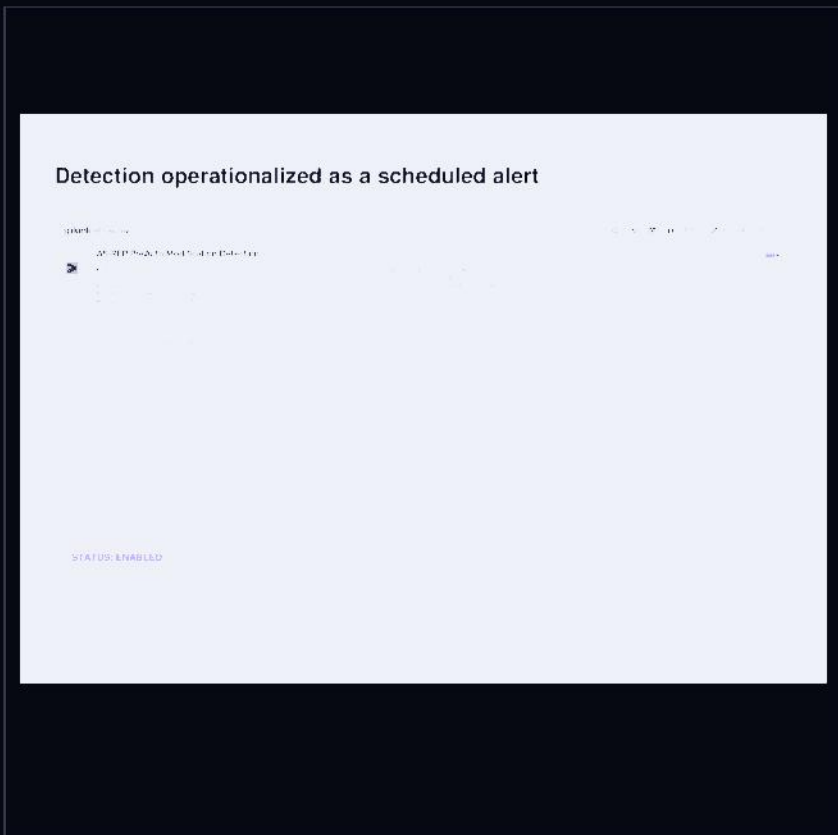
Evidence 1

A focused SPL search was built:



Evidence 2

Additional evidence



This document is a portfolio case-study summary prepared for professional presentation. It documents analysis and detection work performed on training data / lab environments only.